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RESEARCH ARTICLE

Localized dynamic lensing with instantaneous extended depth of field via collinear light–sound interactions

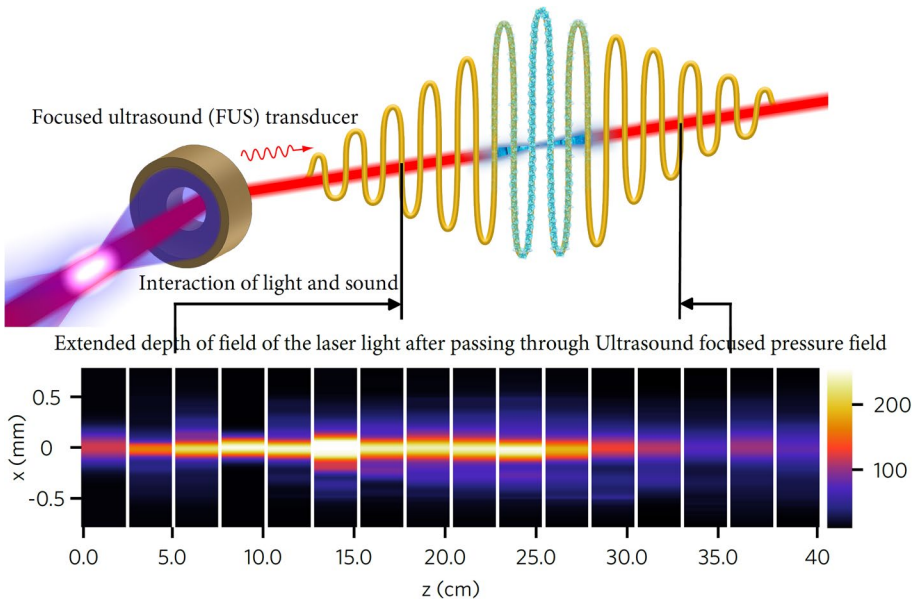
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Abstract

Tunable acousto-optic (AO) lenses have recently emerged as versatile tools for optical beam shaping, imaging, and particle manipulation. Conventional AO lenses rely on light propagating orthogonally to a standing ultrasonic field, producing Bessel-like beam patterns via the Raman–Nath effect that compromise focal localization and depth of field (DoF). Here, we introduce a novel class of AO lenses based on a three-dimensional, dynamically variable refractive index profile generated by a z-axis-scanning focused ultrasound transducer. By exploiting co- and counter-propagating light–sound interactions over an extended axial range, we achieve fully controllable, localized optical focusing with an instantaneous extended DoF. We demonstrate dynamic tuning of focal position, lateral resolution, and optical power throughput by adjusting ultrasound parameters. This approach offers a promising platform for applications requiring precise remote focusing, three-dimensional micromanipulation, and deep tissue imaging or therapy.

Graphical Abstract



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Highlights

- Collinear co- and counter-propagating light–sound interactions create a dynamic 3D-variable refractive index profile for tunable optical focusing.
- The proposed acousto-optic lens achieves localized single-point focus with instantaneous, real-time extended depth of field, without laser–ultrasound synchronization.
- Dynamic control of focal location, depth of field, lateral resolution, and optical throughput is demonstrated by adjusting ultrasound duty cycle and input voltage.
- This novel light–sound interaction scheme shows promise for advanced applications in high-resolution 3D/4D imaging, adaptive microscopy, and remote photonic manipulation.

Keywords Acousto-optic lens · Dynamic focusing · Tunable focal length · Ultrasound modulation · Extended depth of field · Remote optical control

Introduction

Lenses with variable focal lengths are essential in various fields of science and technology, ranging from imaging to sensing and laser beam shaping to microscopy, adaptive optics, and materials processing [1]. Over the years, various techniques have been deployed to realize dynamic focusing, including mechanically moving and micro-electro-mechanical systems devices, electro-optically addressable holographic plates, and acousto-optic (AO) elements. Among them, AO approaches are highly preferred [2, 3] for providing fast switching, adaptability (e.g., in shape and dioptric power), and noninvasiveness with no need for macroscopic moving elements [4].

The AO effect refers to light diffraction due to the variations in refractive index caused by the presence of ultrasound waves [5–7]. Several diffractive AO elements and acoustic gradient-index lenses have been introduced in recent years, generating tunable Bessel-like spots by forming ultrasonic cylindrical standing waves that interact orthogonally with the passing light [3, 4, 8–14] —an effect known as Raman-Nath [15]. By using a confined liquid-filled radial-mode-vibrating piezoelectric shell, these AO lenses enable a combination of tunability, large damage thresholds, and fast z-focus scanning [10, 14]. Such fast z-focusing is accompanied by virtual extended Depth of Field (DoF) owing to collecting multiple focal planes [14]. However, the effective numerical aperture (NA) of these lenses is low—as the aperture is limited by the central lobe of the Bessel function that describes the ultrasound standing wave in the piezo-shell [14]. Also, in addition to generating transversely non-localized (in x–y plane) ring-shaped focal spots, these arrangements tend to suffer from the inability to change the refractive index profile in the z-direction and outside of the AO lens chamber while relying on pulsed laser sources for synchronization. Finally, unlike the ideal Bessel beams that are diffraction-free, the beam, in this case, tends to diffract quickly [16].

Here we present a novel AO dynamic lensing approach that exploits collinear co- or counter-propagating light and focused ultrasound waves to produce a unique, spherically confined refractive index profile for tunable optical focusing. Unlike conventional AO systems that rely on standing-wave Bessel-like fields, our design generates a transversely localized acoustic focal zone to achieve a true single-point optical focus. To realize this, we developed a custom high-frequency z-axis-scan focused ultrasound (FUS) transducer, designed to form a single-point acoustic focus outside and in front of the transducer aperture (Fig. 1b). The thin piezoelectric disk on the front face of the FUS transducer operates in thickness-mode axial vibration (Fig. 2d), producing traveling ultrasonic waves that enhance longitudinal light–sound interaction and induce a 3D-variable refractive index profile (Fig. 2a-b). This configuration enables both laterally localized dynamic focusing and instantaneous extended DoF for continuous-wave (CW) laser beams, without requiring laser–ultrasound synchronization. As illustrated in Fig. 1a, the laser beam is expanded and directed through the water medium to interact with the focused acoustic pressure field generated by the FUS transducer at modest acoustic intensities. Figure 2 schematically depicts the light–sound interaction and induced refractive index gradients that underpin the lensing mechanism.

Modeling and simulation

As an optical beam travels through an acoustic field, variations in the medium's refractive index cause it to deviate from its original path. Depending on the beam's size and the characteristics of the acoustic waveform, this interaction may result in either a deflection or a displacement. Therefore, to model the effect of ultrasonic waves on laser beam deflection, the relationship between variations in the refractive index and acoustic pressure must be derived.

According to Hooke's law, a change in pressure Δp is approximately proportional to a change in density $\Delta\rho$ as the following:

$$\Delta p = \frac{B_w}{\rho_0}\Delta\rho \tag{1}$$

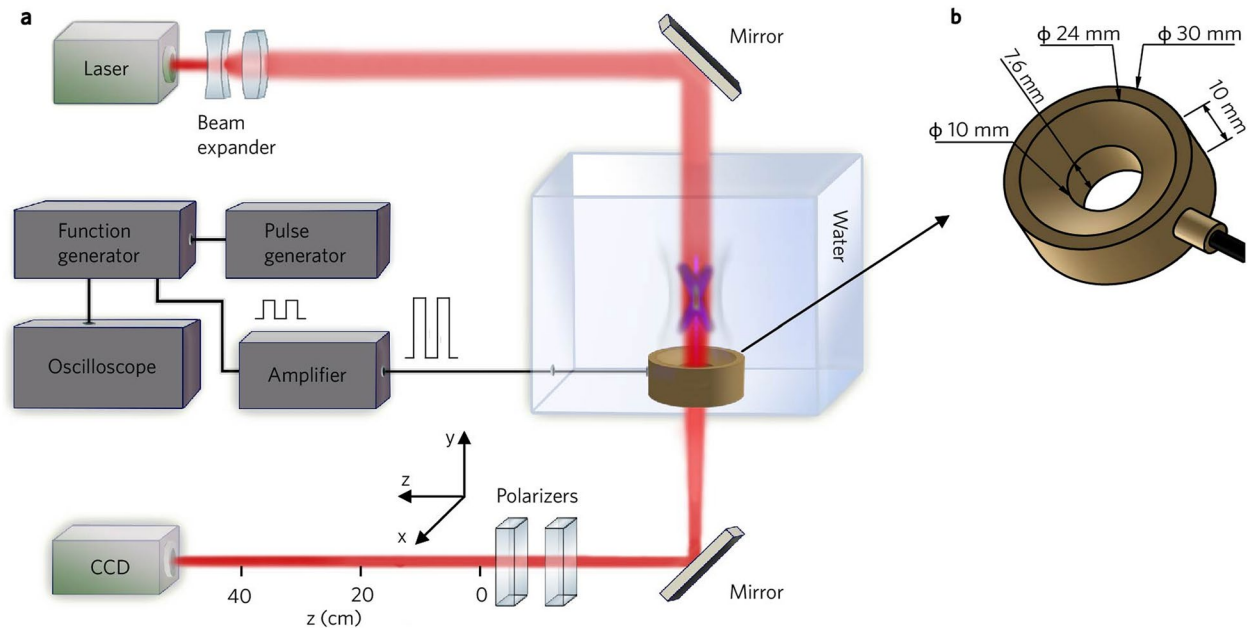


Fig. 1 **a** A schematic of the experimental setup for counter-propagating alignment (see Supplementary materials 1 and 2 for co-propagating setup). **b** A schematic of the forward-scan ring focused ultrasound (FUS) transducer

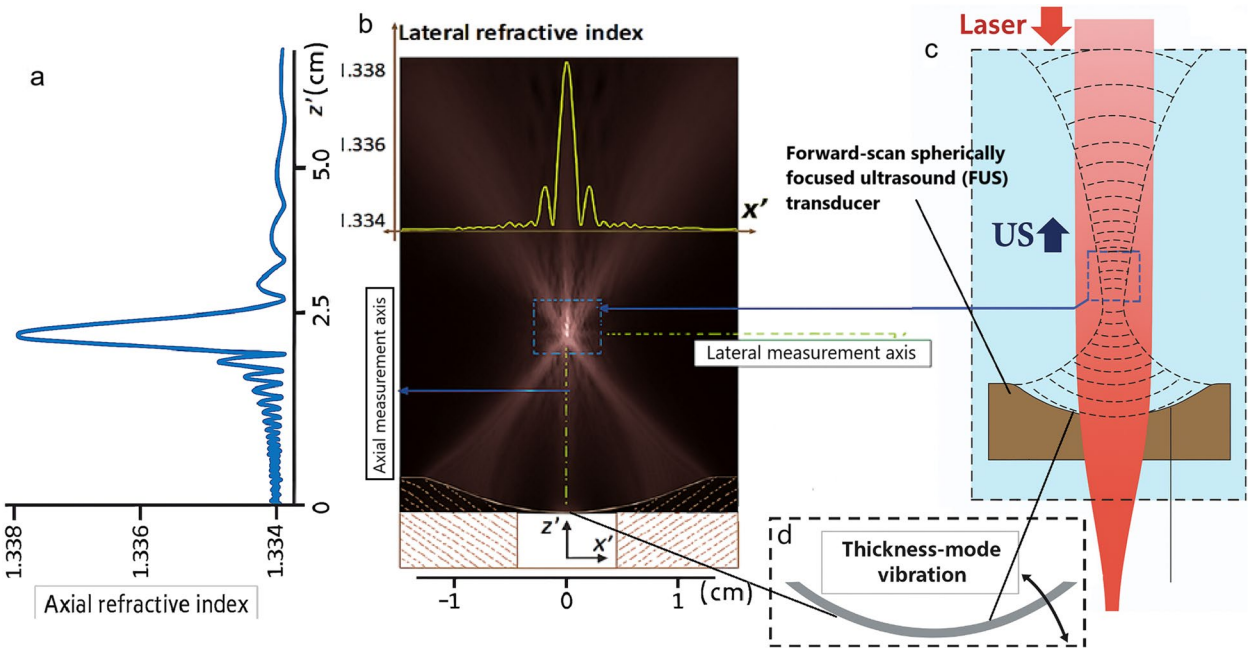


Fig. 2 **a** Induced axial refractive index profile. **b** Simulated ultrasonic pressure field generated by the FUS transducer (shown in dark brown), which induces a 3D refractive index contrast in both the lateral (yellow plot) and axial (blue plot, shown in part a) directions. **c** Schematic illustration of the interaction between the counter-propagating laser beam and the ultrasound (US) pressure field. **d** Schematic of the piezoelectric element in the FUS transducer illustrating thickness-mode vibration

where p is the acoustic pressure, ρ_0 is the ambient density and B_w is the bulk modulus of elasticity, respectively.

The International Association for the Properties of Water and Steam (IAPWS) developed a model that expresses the refractive index of a medium as a function of wavelength, temperature, and pressure as the following [17]:

$$n = \sqrt{\frac{1 + 2(1+\overline{p})A}{1 - (1+\overline{p})A}} \tag{2}$$

where n is the refractive index, $\overline{p}$ is the normalized pressure ($\frac{p}{B_w}$) and the parameter A is a function of optical wavelength, temperature and density as:

$$A = a_1\overline{\rho} + a_7\overline{\rho}^2 + a_8\left(\overline{T}, \overline{\lambda}\right) = a_9 + a_{10}\overline{p} + a_7\overline{p}^2 \tag{3}$$

where $\overline{\rho}$ is normalized density ($\frac{\rho}{\rho_0}$), $\overline{T}$ is the dimensionless temperature, $\overline{\lambda}$ is wavelength, and a_i's are empirical constants listed in Table 1.

A simplified version the refractive index can be expressed as:

$$n = \sqrt{\frac{a_{14} + 2\chi(\overline{p})}{a_{15} - \chi(\overline{p})}} \tag{4}$$

$$\chi(\overline{p}) = a_{11}\overline{p} + a_{12}\overline{p}^2 + a_7\overline{p}^3 \tag{5}$$

When light passes through a medium with a dynamic refractive index, it bends by an angle θ_b. The amount of beam bending is calculated using the following equation:

$$\theta_b = \int \frac{1}{n(x,z)} \frac{\partial n(x,z)}{\partial x} dz \tag{6}$$

For simulating the problem under investigation, the pressure field of the transducer was first modeled using COMSOL Multiphysics software. The details of simulation are omitted here for the sake of brevity. Figure 3 illustrates the modulated refractive index induced by ultrasounds. The range of refractive index variation is from 1.334 to 1.338, which corresponds to approximately 0.3%. As evident from the image, the highest refractive index is located at the focal point.

Following the calculation of the refractive index, the light propagation in a medium with modulated refractive index induced by ultrasounds is simulated using Beam Propagation Method [18] with the split-step algorithm in MATLAB. The light source is assumed to be a Gaussian beam with 633 nm wavelength and 6.12 mm beam width. Figure 4 shows the simulated laser beam propagation in water (Fig. 4b) and free space (Fig. 4c), calculated using the axial intensity definition in Fig. 4a. These simulations use the modulated refractive-index distribution from Fig. 3. As seen in Fig. 4, the light intensity increases near the acoustic focal region and remains extended along the longitudina (z) direction, with symmetric profiles in the x and y directions. In free-space environment, the light intensity reaches its maximum at z = 122.67 µm. The light intensity profile along the center and in the longitudinal direction is shown in Fig. 5. According to this figure, the DoF of the modulated laser beam is found to be approximately 49 cm (~0.5 m).

Figure 6 presents the light-intensity distribution in the transverse plane at the focal point (z = 122.67 mm) for two conditions: with the ultrasonic transducer turned OFF (unmodulated laser) and ON (modulated laser). When the FUS is ON, the ultrasound-induced refractive-index grating produces strong lateral focuing, resulting in a significantly narrower and more intense beam profile at the focal plane.

Table 1 Constants and derived quantities

$a_0 = 0.244257733$	$a_1 = 9.74634476 \times 10^{-3}$
$a_2 = -3.73234996 \times 10^{-3}$	$a_3 = 2.65666426 \times 10^{-4}$
$a_4 = 2.45733798 \times 10^{-3}$	$a_5 = 2.45934259 \times 10^{-3}$
$a_6 = 0.900704920$	$a_7 = -1.66626219 \times 10^{-2}$
$\lambda_{UV} = 0.229202$	$\lambda_{IR} = 5.432937$
$a_8 \equiv a_0 + a_2\bar{T} + a_3\bar{\lambda}^2\bar{T} + \frac{a_4}{\bar{\lambda}^2} + \frac{a_5}{\bar{\lambda}^2 - \bar{\lambda}_{UV}^2} + \frac{a_6}{\bar{\lambda}^2 - \bar{\lambda}_{IR}^2}$	
$a_9 = a_1 + a8 + a_7$	$a_{10} = a_1 + 2{*}a_7$
$a_{11} = a_9 + a_{10}$	$a_{12} = a_7 + a_{10}$
$a_{14} = 1 + 2{*}a_9$	$a_{15} = 1 - a_9$

Experimental setup

The FUS transducer has an aperture size of 24 mm, with an acoustic focal length of 23 mm (in front of it). It was designed and fabricated with PZT-8 (a high mechanical-Q material) and a final thickness of 0.46 mm. The electrical circuit, as well as matching and backing layers (made from epoxy) were placed inside a custom-machined brass housing. The average center frequency of the transducer was 3.3 MHz, and the −6-dB bandwidth was 51%, as estimated from the measured pulse-echo response. The average insertion loss was measured to be 16.3 dB.

In the characterization setup, the Gaussian beam from a He–Ne class-II laser (Oriel Corporation) with an output power of 0.95 mW, centered at a wavelength of 633.2 nm, was expanded, and collimated with a beam expander to a beam diameter of 6.12 mm. The laser was direct to a raised fish tank filled halfway with deionized milli-pure water. The FUS transducer was submerged in the water upward with the focal face pointing away from the bottom of the tank (for counter-propagating setup, Figs. 1 and supplementary material 3). FUS transducer was arranged either in a co-propagating configuration (Fig. S1) or a counter-propagating configuration (Fig. 1 and supplementary material 3) relative to the laser-beam propagation, with the beam entering through the bottom of the fish tank. A photograph of the experimental setup is shown in supplementary material 3, showing the tank, FUS transducer, laser source, CCD camera, and related equipment. Table 2 summarizes the key characteristics of the experimental setup.

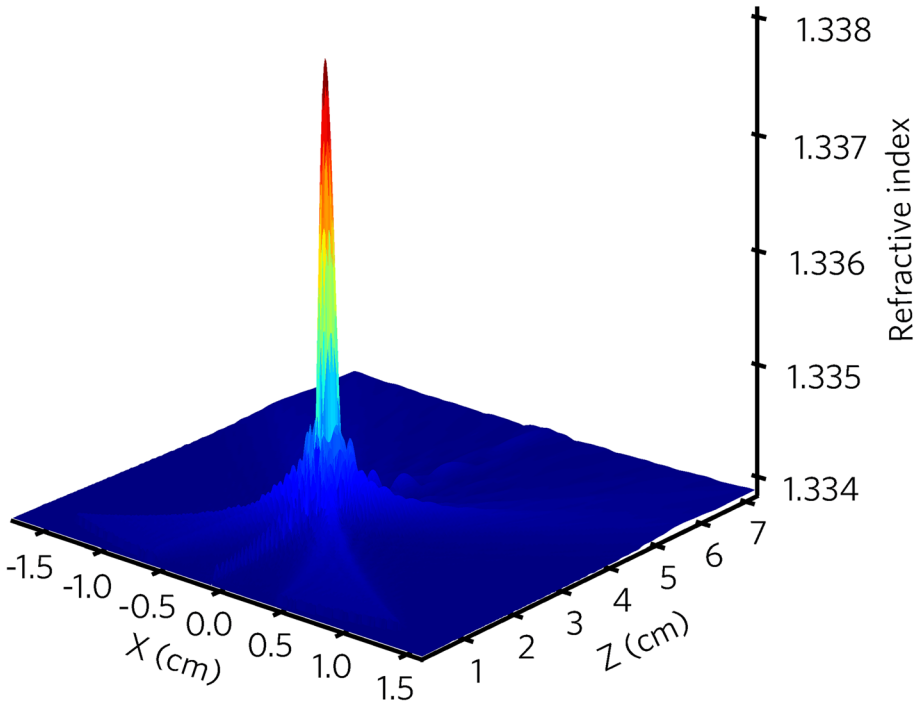


Fig. 3 The modulated refractive index induced by ultrasound

Experimental results and discussion

Figure 7 presents the 2D intensity profiles of the light beam, recorded by the CCD camera at FUS input duty cycles of 2% and 20% (i.e., the fraction of each period during which the ultrasound signal is active). These profiles reveal a localized focus (single-point focus in the x–y plane) when the FUS transducer is excited. The duty cycle plays a critical role in shaping the laser beam: at a 2% duty cycle, the beam width is significantly narrower compared to the 20% case, making the 2% duty

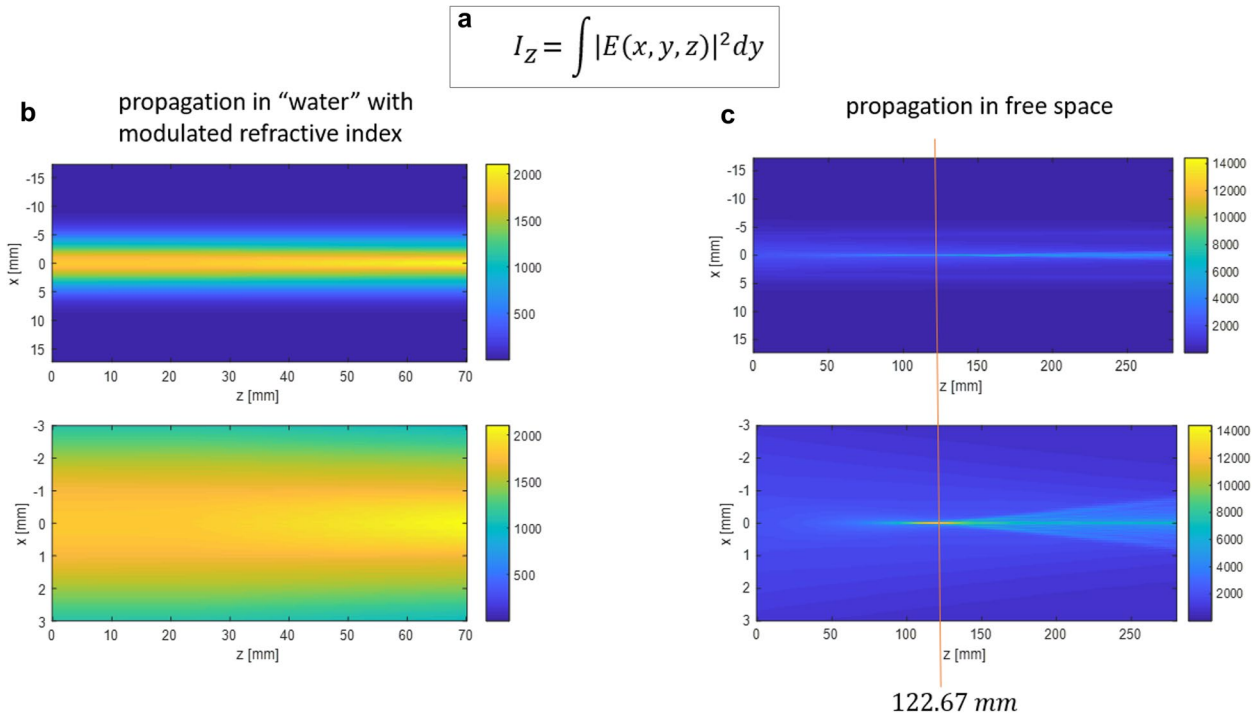


Fig. 4 Simulated intensity profiles of the modulated light along the propagational direction in water (**b**) and in free space (**c**), calculated using the intensity expression shown in (**a**)

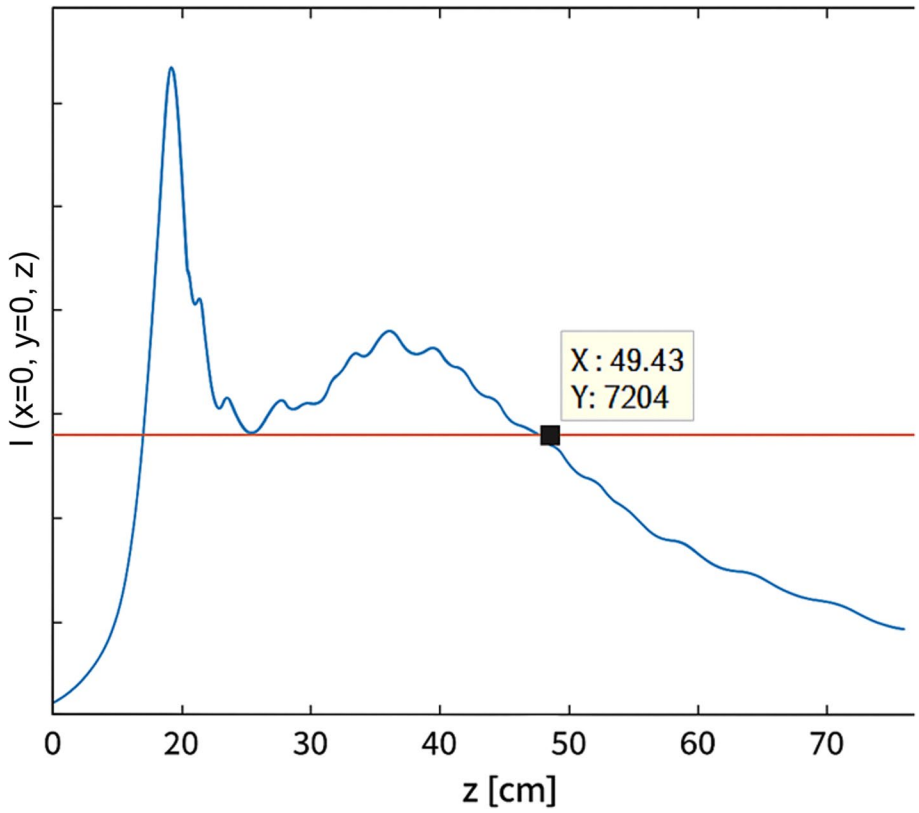


Fig. 5 Simulated intensity profiles of the modulated light in free space

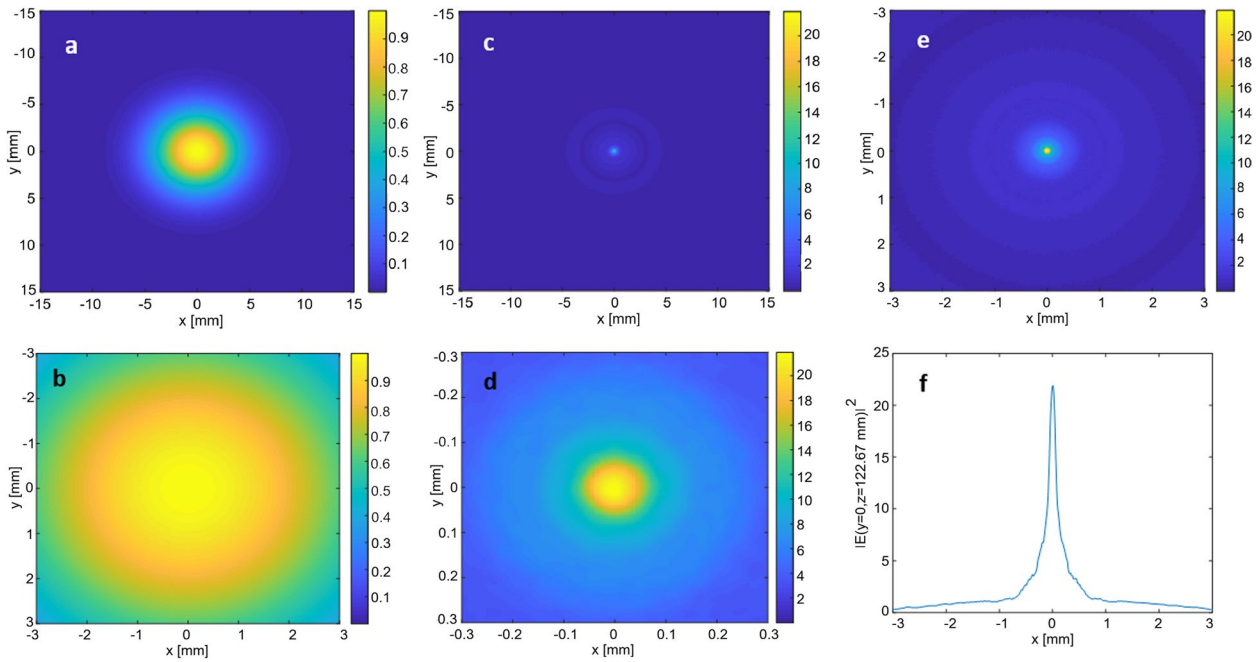


Fig. 6 Simulated x-y light intensity cross-section profile at z = 122.67 mm in free space (focal plane). **a**,**b** FUS OFF (no refractive-index modulation), **c**-**e** FUS ON. showing modulation by the refractive-index grating from a spherically fucused ultrasound wave. **f** 1D intensity cut along x at y = 0 and z = 122.67 mm, $|E$ (y = 0, z = 122.67 mm$|^2$, highlighting the sharper beam confinement with FUS.

cycle more suitable for high-resolution imaging applications. Conversely, a 20% duty cycle can be advantageous when higher optical intensity and greater light throughput are desired.

In these experiments, the FUS transducer was driven at 84 V. At duty cycles of 2% and 20%, the x–y cross-sectional beam area was reduced by factors of approximately 477 (Fig. 7b compared to a and c) and 208 (Fig. 7e compared to d and f), respectively, corresponding to equivalent increases in power density. These factors were calculated by comparing the pre-FUS beam area (approximately 3.61 mm × 3.62 mm) to the during-FUS full-width at half maximum (FWHM) (approximately 0.16 mm × 0.17 mm) at 2%, and 3.63 mm × 3.64 mm to 0.24 mm × 0.26 mm at 20%. These results indicate that the power density enhancement at a 2% duty cycle is more than twice that achieved at 20%.

Figure 8 shows the variation of lensing power over time for two different duty cycles, indicating a relatively stable behavior.

In addition to the reduction in laser beam width and its increased focusing, the extension of the DoF is one of the key features of the ultrasonic interaction with the laser beam. The dynamic control of focal properties in our system arises from the interaction between the FUS pressure field and the optical beam. The acoustic pressure amplitude (P_{ac}) is approximately proportional to both the input voltage (V) and the duty cycle (D), since higher duty cycles deliver more average power to the medium. This acoustic field induces a spatially varying refractive index contrast in the medium, governed by the relation $\Delta n = C \cdot P_{ac}$, where C is the acousto-optic coefficient. A larger Δn corresponds to a stronger refractive index gradient, which bends light more sharply, leading to a shorter optical focal length and a narrower DoF. This relationship can be summarized approximately by the following proportionalities:

$$f_{optical} \propto 1/\Delta n(V, D) \propto 1/(V \cdot D), \text{ and DoF } \propto f_{optical}$$

Figures 9 and 10 illustrate the experimental results validating these principles. Figure 9 presents measurements of optical beam diameter and FWHM in the x and y directions along the propagation axis for two different FUS duty cycles (2% and 20%), both at a constant voltage of 84 V. Panel (a) corresponds to the 2% duty cycle and (b) to the 20% duty cycle. Panels (a,b) compare the depth of focus (DoF), which is approximately 28 cm for 2% and 11.5 cm for 20%. During-FUS results (green) show significantly narrowed beam diameters and FWHM compared to the unmodulated cases (brown). These results confirm that lower duty cycles produce longer instantaneous DoF with sharper focus.

By stating that the extended DoF is instantaneous, we mean that the extended optical focus appears physically and in real time—unlike in TAG lens systems, where it is achieved by rapidly scanning multiple focal planes and integrating them into a single image, thereby creating a virtually extended DoF. In contrast, our C2AM system generates a true, static extended DoF without scanning, due to the collinear alignment and interaction of the focused ultrasound and laser beams. We refer to this as instantaneous DoF. Further investigation is ongoing to fully understand the mechanism behind this behavior.

Figure 10 shows the effect of varying the FUS input voltage on the axial location of the optical focal spot. At the maximum input voltage of 105 V (panel a, duty cycle 5%), the focal point is observed at z = 4 cm. As the voltage decreases to 84 V (panel b, duty cycle 2%) and 63 V (panel c, duty cycle 20%), the focal spot shifts progressively farther away to z = 8 cm and z = 10 cm, respectively. Despite the differences in duty cycle across these conditions, the results indicate that the input voltage has a more pronounced effect on the axial focal position than the duty cycle. The relationship among focal length, voltage, and duty cycle appears nonlinear, likely due to complex interactions between acoustic pressure dynamics and medium response under varying power levels.

Table 2 The characteristics of experimental setup

FUS Transducer (A custom-made z-axis-scan FUS)	
Focal length/Aperture size	23 mm/24 mm
Center frequency/Bandwidth	3.3 MHz/51%
Piezoelectric Material/Thickness	PZT-8/0.46 mm
FUS acoustic intensity	119 W/cm^2
Insertion loss	16.3 dB
Laser (He–Ne class-II laser, Oriel Corporation)	
Output power	0.95 mW
Wavelength	633.2 nm
Laser beam diameter	6.12 mm
Camera (EM-CCD, Hamamatsu, C9100-13)	
Frames per second	31.9 fps

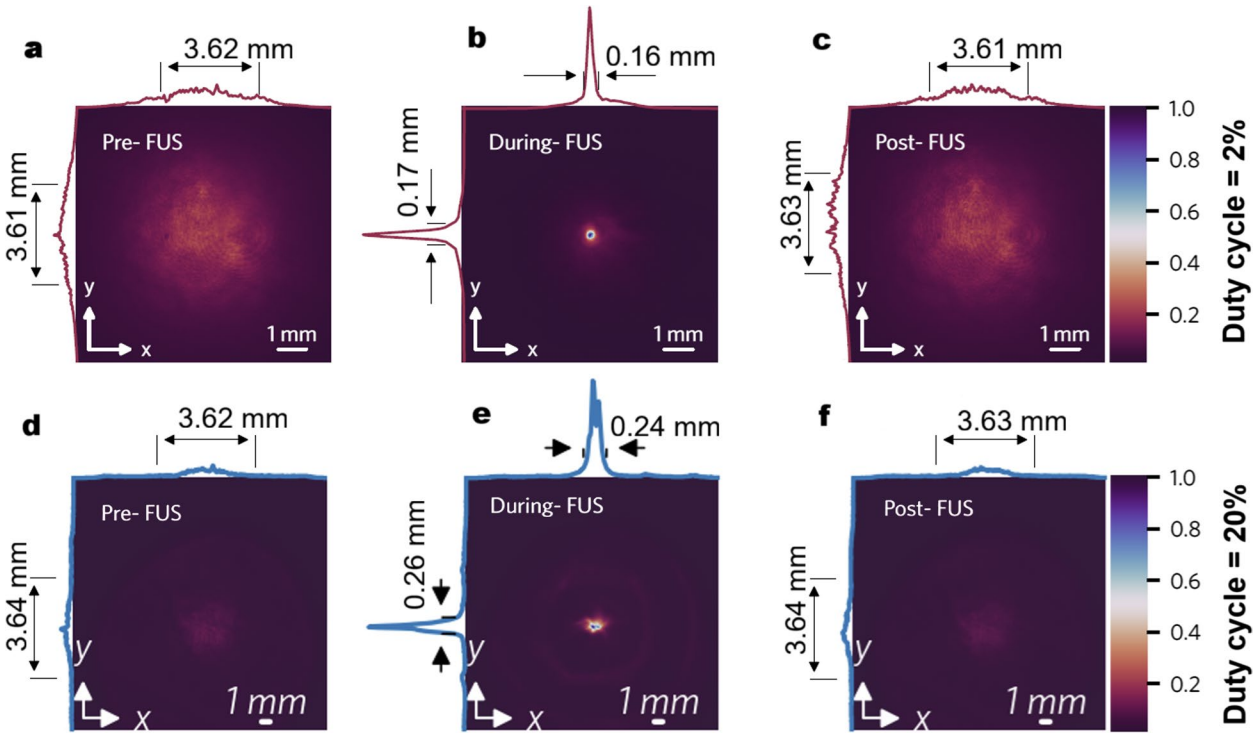


Fig. 7 Observed 2D light intensity patterns and FWHM recorded by the CCD camera in pre- (**a**, **d**), during- (**b**, **e**), and post- (**c**, **f**) FUS states. FUS is ON during (**b**, **e**) and OFF in pre- (**a**, **d**) and post- (**c**, **f**) states, shown for two different duty cycles

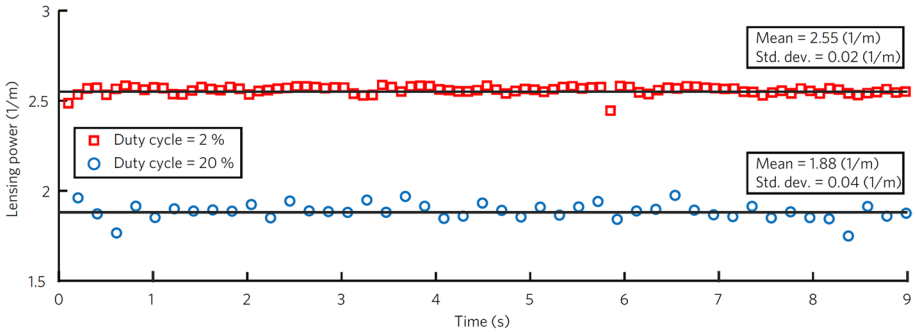


Fig. 8 Distribution of the measured lensing power over the time for two extreme duty cycles of 2% (red squares) and 20% (blue circle), respectively

Our experimental results demonstrate dynamic and tunable control of key optical parameters—focal point location, DoF, lateral resolution, FWHM, and power throughput—by adjusting the FUS input voltage and duty cycle. Figures 7 and 9 show that lower duty cycles (e.g., 2%) at 84 V yield a sharper focus, narrower FWHM, higher lateral resolution, and longer DoF, while higher duty cycles (e.g., 20%) deliver greater optical power but at the cost of reduced resolution and shorter DoF. Figure 10 further confirms that the axial focal position is primarily governed by the input voltage rather than duty cycle. Together, these results demonstrate that our system enables flexible and precise tuning of optical focusing performance, balancing resolution, DoF, focal location, and power delivery to meet specific application needs.

Figure 11 illustrates the measured light intensity profiles along the propagation axis (z) for various FUS input voltages and duty cycles, demonstrating how these parameters guide and modulate the localized focus within the extended DoF. As shown, increasing the voltage or adjusting the duty cycle shifts the axial location of peak intensity, allowing precise control of the focal position while maintaining a confined beam.

Table 3 summarizes the measured characteristics of the output light beam for two FUS duty cycles at a constant input voltage of 84 V. As shown, the lower duty cycle (2%) yields

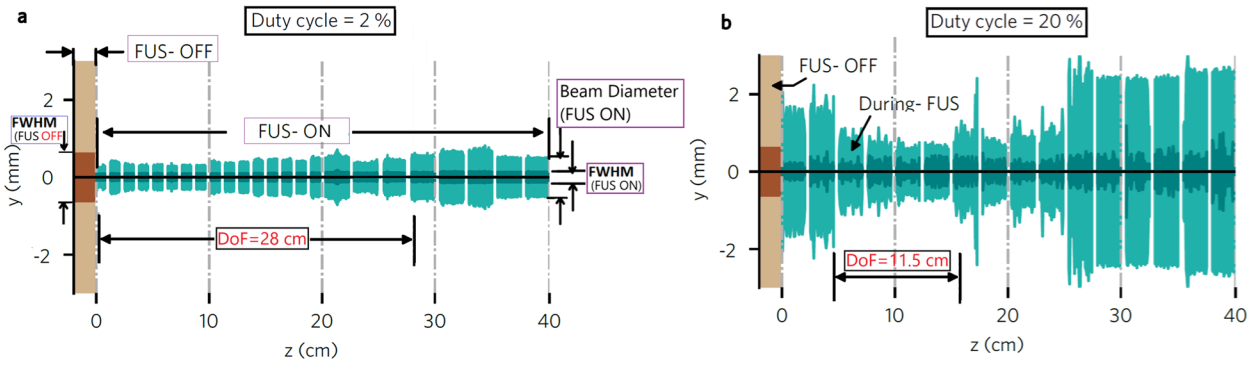


Fig. 9 Experimental results of optical beam diameter and FWHM in the x and y directions along the propagation axis (z) at FUS input duty cycles of 2% (**a**) and 20% (**b**), both at an input voltage of 84 V. The measured DoF is approximately 28 cm for the 2% duty cycle and 11.5 cm for the 20% duty cycle, as indicated. The unmodulated (FUS-OFF) beam diameters and FWHM in x and y are shown at z = 0 in brown, while the narrower beam profiles during FUS-ON are annotated in the figures, demonstrating the focusing effect and extended DoF under different duty cycles

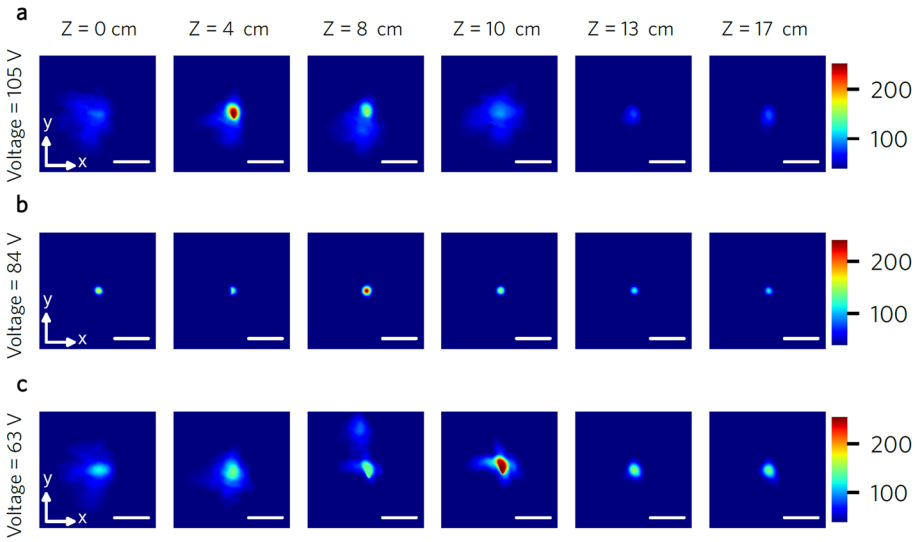


Fig. 10 Dynamic control of optical focusing along the propagation axis. Panels (**a**–**c**) show CCD camera-recorded intensity patterns at different FUS input voltages. As the input voltage decreases, the optical focal spot shifts progressively farther from the C2AM lens. The scale bar is 1 mm

a significantly narrower FWHM and a longer depth of field, demonstrating a more localized and elongated optical focus.

To compare our C2AM system with conventional AO lenses, particularly Tunable Acoustic Gradient (TAG) lenses, we evaluated both the effective NA and refractive index modulation depth (Δn). The NA in our system is estimated from the measured FWHM of the optical focus using the relation $NA \approx 0.51 \times \lambda/FWHM$, where $\lambda = 633.2$ nm. With a minimum measured FWHM of 0.16 mm, we obtain NA ≈ 0.0078, which is significantly higher than the typical range reported for TAG lenses in the literature (NA ~ 0.0005–0.001) [9]. The improved confinement in our system arises from the traveling-wave focused ultrasound field, which is not limited by the central lobe size inherent to the standing-wave Bessel beam in TAG lenses [14].

Furthermore, the refractive index modulation depth in our system is estimated as $\Delta n = C \cdot P_{peak}$ [2], where $C \approx 1.3 \times 10^{-10}$ Pa^{-1} is the AO coefficient of water. Using the relation $I = (P_{rms})^2/(\rho c)$, the root-mean-square pressure is calculated as $P_{rms} \approx 19.36$ MPa, giving P_peak ≈ 27.4 MPa at 105 V, which yields $\Delta n \approx 3.56 \times 10^{-3}$. This is at least an order of magnitude greater than TAG lenses, which typically achieve Δn on the order of 10^{-5}–10^{-4} [9, 13].

The ability to steepen the refractive index gradient and achieve shorter optical focal lengths can be addressed through several approaches. Increasing the ultrasound frequency f decreases the acoustic wavelength $\lambda_s = c_s/f$ and thereby narrows the lateral beam width $W_b \approx f_\# \times \lambda$ and depth of focus $D_f \approx f_\#^2 \times \lambda$. As optical focusing in our system is governed by the steepness of the induced refractive index profile, higher ultrasound frequencies produce a more tightly focused acoustic beam with sharper pressure and refractive index variations, enabling stronger optical focusing. Additionally, replacing water ($C \approx 1.3 \times 10^{-10}$ Pa^{-1}) with a medium of higher

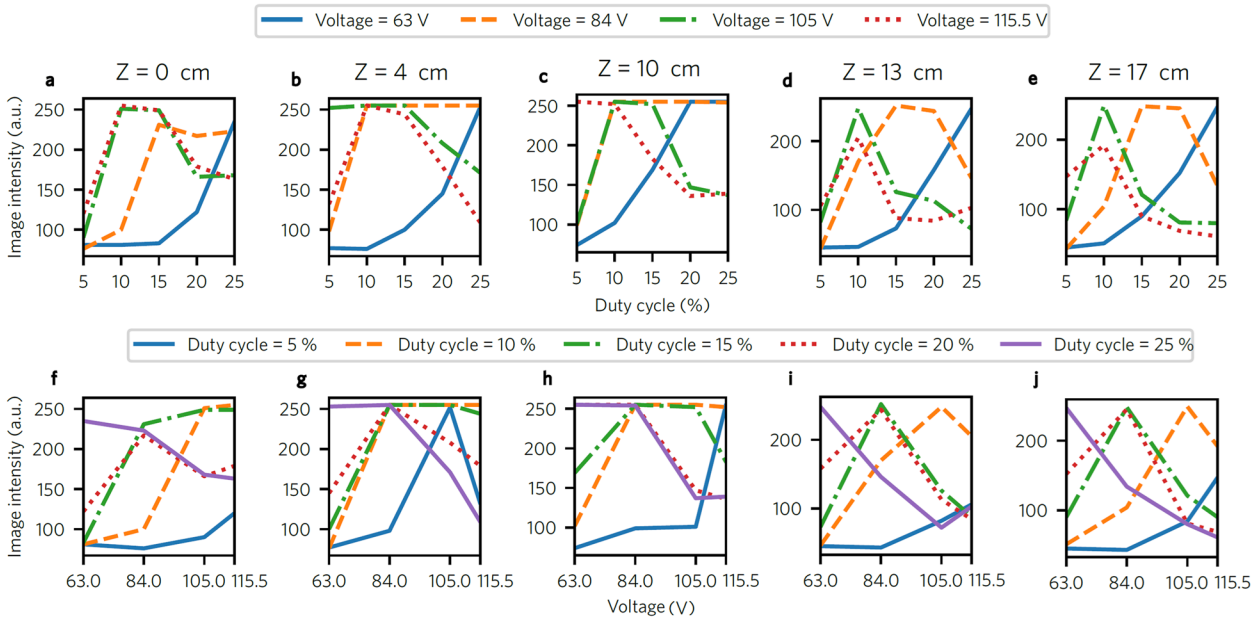


Fig. 11 Measured modulation of light intensity by ultrasound input voltage and duty cycle over an extended localized DoF. Experimental results show light intensity variations as a function of FUS input voltage (**a**–**e**) and duty cycle (**f**–**j**) at different positions along the propagation axis (z). In panels f–j, peak intensities corresponding to each voltage (**a**–**e**) and duty cycle (**f**–**j**) are observed at distinct axial locations

Table 3 The measured characteristics for the output light beam

	FUS Duty cycle	
	2%	20%
FWHM (mm × mm)	0.16 × 0.17	0.24 × 0.26
FWHM reduction	477- fold	218- fold
Depth of field (cm)	11.5	28
Lensing power (1/m)	2.55	1.88

*For FUS input voltage = 84 V

photoelastic coefficient—such as glycerol–water mixtures or silicone oils—can increase Δn at the same acoustic pressure, thereby producing a stronger gradient. Other strategies include optimizing the ultrasound field through phased arrays or nonlinear focusing techniques to enhance confinement.

It is worth noting that the acoustic wave frequency, the size of the piezoelectric element, and the sound velocity in the medium influence the lens switching speed [9]. The thin piezoelectric material in our z-axis-scan FUS transducer, which supports higher resonance frequencies, is expected to enable faster focal switching than existing AO elements [10]. Potentially, up to two orders of magnitude higher switching speed could be achieved using even thinner disks. Furthermore, the electromechanical coupling coefficient of our transducer is nominally higher than that of cylindrical transducers commonly used in AO lenses, favoring lower insertion loss, wider frequency response, improved axial resolution, and higher acoustic penetration [19–21].

Our results demonstrate that the maximum applied intensity (25 kW/cm^2) at 105 V remains below the cavitation threshold in air-free water [22], avoiding cavitation and Mie scattering effects. This ensures that the observed optical focusing performance in our experiments is not compromised by bubble formation.

To enable future translation of our approach to deep tissue imaging, it is important to consider the potential for cavitation, particularly when using pulsed laser irradiation, which achieves higher peak powers but also increases the risk of cavitation due to rapid thermal expansion and acoustic transients. Our preliminary simulations and tissue experiments indicate that CW laser operation enables improved optical focusing at ultrasound frequencies above 2–3 MHz, while keeping cavitation risk minimal. Nevertheless, the cavitation threshold is expected to vary depending on tissue properties and operating parameters, and further investigation is warranted. In addition, our system offers flexibility to mitigate cavitation risks and improve portability. Redesigning the FUS transducers—for example, using smaller apertures, lower voltage drives, or alternative frequency ranges—can reduce acoustic intensities while maintaining sufficient refractive index modulation.

We also plan to explore alternative coupling media with higher elasto-optic coefficients and higher refractive indices, including both liquids and solids, which could enhance lensing efficiency and allow for a smaller, more compact container. Since our FUS transducers are already compact and fabricated in-house, further miniaturization and optimization

are feasible and planned as part of our ongoing work to develop a portable, tissue-compatible system.

In earlier studies [16], the light and ultrasound paths were either perpendicular, as in the Raman-Nath scattering regime ([8], foundational to TAG lens technology), or at a small Bragg angle, still near-perpendicular. In our configuration, the Bragg angle for first-order diffraction can be calculated as follows. Assuming a light wavelength in vacuum $\lambda_0 = 633$ nm, refractive index of water $n_m \approx 1.33$, ultrasound frequency $f_s = 3.3$ MHz, and speed of sound $c_s \approx 1500$ m/s, the light wavelength in water is $\lambda = \lambda_0/n_m \approx 476$ nm, and the acoustic wavelength is $\Lambda = c_s/f_s \approx 454.5$ µm. Using Bragg's law: $n\lambda = 2\Lambda \cdot \sin\theta$. For first-order diffraction ($n = 1$): $\sin\theta \approx \lambda/(2\Lambda) \approx 5.237 \times 10^{-4}$, yields $\theta \approx 0.03°$. This calculation confirms that the angle between ultrasound and light in prior configurations is close to 90°, while our system operates in a truly co-linear configuration with ultrasound and laser beams either co-propagating or counter-propagating.

This co- and counter-propagating configuration enables distinct focal behaviors and enhanced longitudinal interaction. In our experiments, the counter-propagating setup consistently produced a more stable, sharply defined focus with higher resolution, likely due to asymmetries in tank alignment or optical path, which we plan to address in future compact setups. Additionally, mid-range interaction angles may offer a balance between stability and confinement, which we intend to explore using phased arrays or adjustable transducer orientations.

Conclusion

We have demonstrated a novel class of dynamic AO lenses based on co-linear interaction between a laser beam and a focused traveling ultrasound field (C2AM system). This approach enables a physically localized optical focus in the lateral plane and an extended DoF along the propagation axis, realized instantaneously without scanning or computational reconstruction. By tuning the ultrasound input voltage and duty cycle, we achieved flexible, precise control of key optical parameters—including focal point location, DoF, lateral resolution, FWHM, and optical power throughput. Lower duty cycles favored sharper focus and longer DoF, whereas higher duty cycles enhanced optical power at the expense of resolution. The axial focal position was primarily governed by input voltage, enabling additional tunability.

The co- and counter-propagating geometries distinguish this system from conventional AO lenses, offering higher index modulation, tighter confinement, and broad tunability. These features make C2AM a promising platform for imaging, manipulation, and photonic applications, while laying the groundwork for future compact and tissue-compatible implementations.

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Declarations

Ethics approval and consent to participate Not applicable.

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